

Project Report On Automated Employee Onboarding & Offboarding System in ServiceNow

Project Title: Automated Employee Onboarding & Offboarding System in ServiceNow

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GitHub Repository: <https://github.com/Priyahoskote/ServiceNow>

Onboarding and Offboarding



Abstract

This project demonstrates the automation of employee onboarding and offboarding processes in ServiceNow using Service Catalog, Catalog Variables, Flow Designer, Approvals, and Notifications.

The system enables users (HR or employees) to raise onboarding or offboarding requests through a catalog item. It captures all required employee details and processes the request automatically using Flow Designer. The system dynamically assigns tasks to IT, Facilities, and Security teams, ensures approval from the manager, and updates the request status based on task completion.

This solution eliminates manual coordination, reduces delays, and improves visibility across departments by automating the entire employee lifecycle workflow.

Introduction

ServiceNow is a cloud-based IT Service Management (ITSM) platform that enables organizations to automate business processes efficiently. One of its most powerful features is workflow automation using Flow Designer.

Employee onboarding and offboarding are critical processes in organizations. When handled manually, they often lead to delays, miscommunication, and missed responsibilities between departments.

This project addresses these issues by:

1. Creating a Service Catalog item for onboarding/offboarding requests
2. Automating approvals using Flow Designer
3. Assigning tasks to multiple departments
4. Tracking request progress in real time
5. Sending automated notifications

Objectives

1. **Automation** – Eliminate manual handling of onboarding/offboarding
2. **Standardization** – Ensure structured workflow execution
3. **Efficiency** – Reduce delays in approvals and task handling
4. **User Experience** – Provide a dynamic and simple request form
5. **Transparency** – Enable real-time tracking of request status in real time.

Literature Review

This project is built using key ServiceNow concepts:

- Service Catalog – Interface for users to submit requests
- Catalog Variables – Capture employee details
- Catalog UI Policies – Dynamic form behavior
- Flow Designer – Automates the workflow
- Approvals – Ensures authorization by manager
- Catalog Tasks (sc_task) – Assigns work to departments
- Notifications – Sends email updates

This project integrates all these components into a unified automated system.

Methodology

The project follows a step-by-step approach:

Step 1: Service Catalog Item Creation

A catalog item named:

“Employee Request”

was created to capture employee onboarding/offboarding requests.

Catalog Item
Employee Request

Build and modify items faster with the improved [Catalog Builder](#).

NameEmployee Request

ApplicationGlobal

Active

Roles

Fulfillment automation levelUnspecified

CatalogsService Catalog

CategoryServices

State-- None --

Checked out-- None --

OwnerSystem Administrator

Item Details

Process Engine

Picture

Pricing

Portal Settings

Short description

Description

Meta

Copy

Try It

Update

Edit in Catalog Builder

Delete

Step 2: Defining Variables and Variable Sets

The catalog item includes fields such as:

- Employee ID
- Requested For
- Manager
- Department
- Joining Date / Exit Date
- Assets Needed / Assets to Collect
- Request Type (Onboarding / Offboarding)

Item Diagnostic

Run Point Scan

[SN Utils] Versions (28)

Variables (11)

Variable Sets

Catalog UI Policies (2)

Catalog Client Scripts

Available For

Not Available For

Categories (1)

Catalogs (1)

Catalog Data Lookup Definitions

Related Articles

Related Catalog Items

Assigned Topics

Order

Search

Actions on selected rows...

New

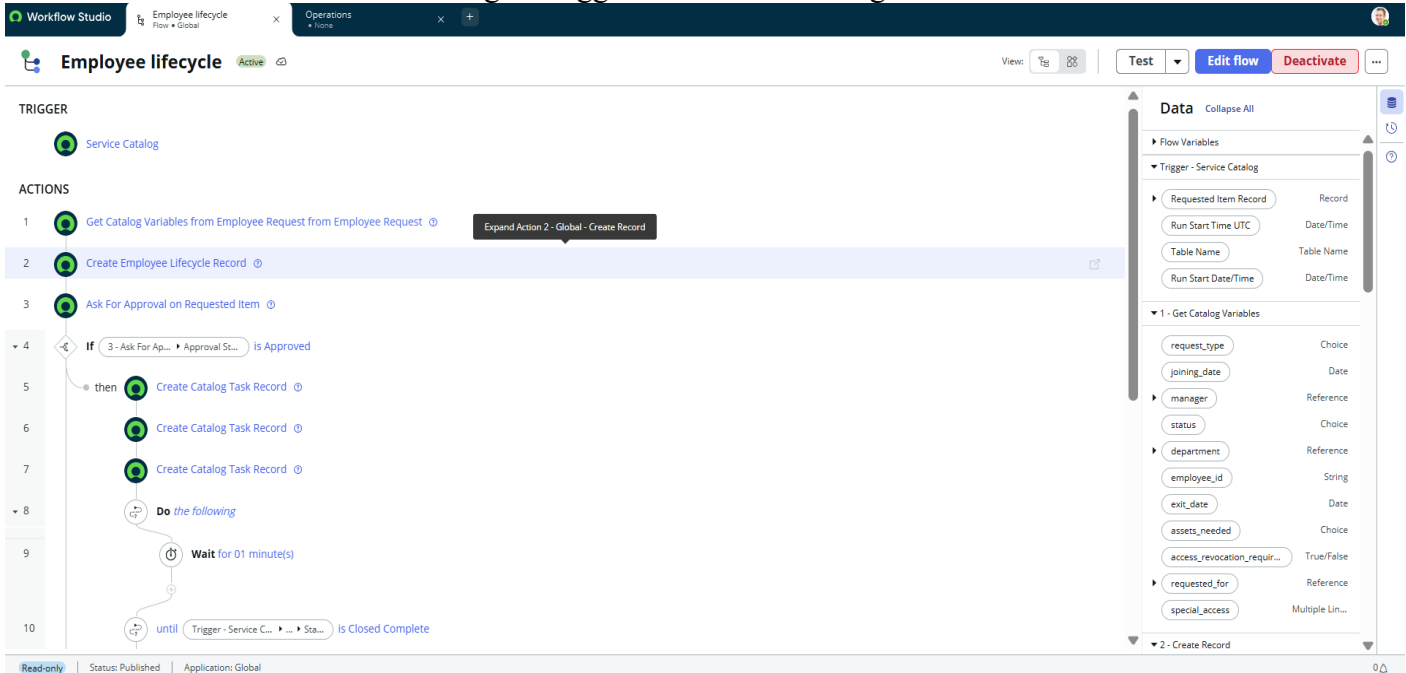
Catalog item = Employee Request

Type	Question	Order
Yes / No	access revocation required	
Select Box	Request Type	
Reference	Requested For	
Multi Line Text	special access	
Date	Joining Date	
Reference	Manager	
Multiple Choice	Status	
Reference	Department	
Single Line Text	Employee ID	
Date	Exit date	
Multiple Choice	Assets needed	

1 to 11 of 11

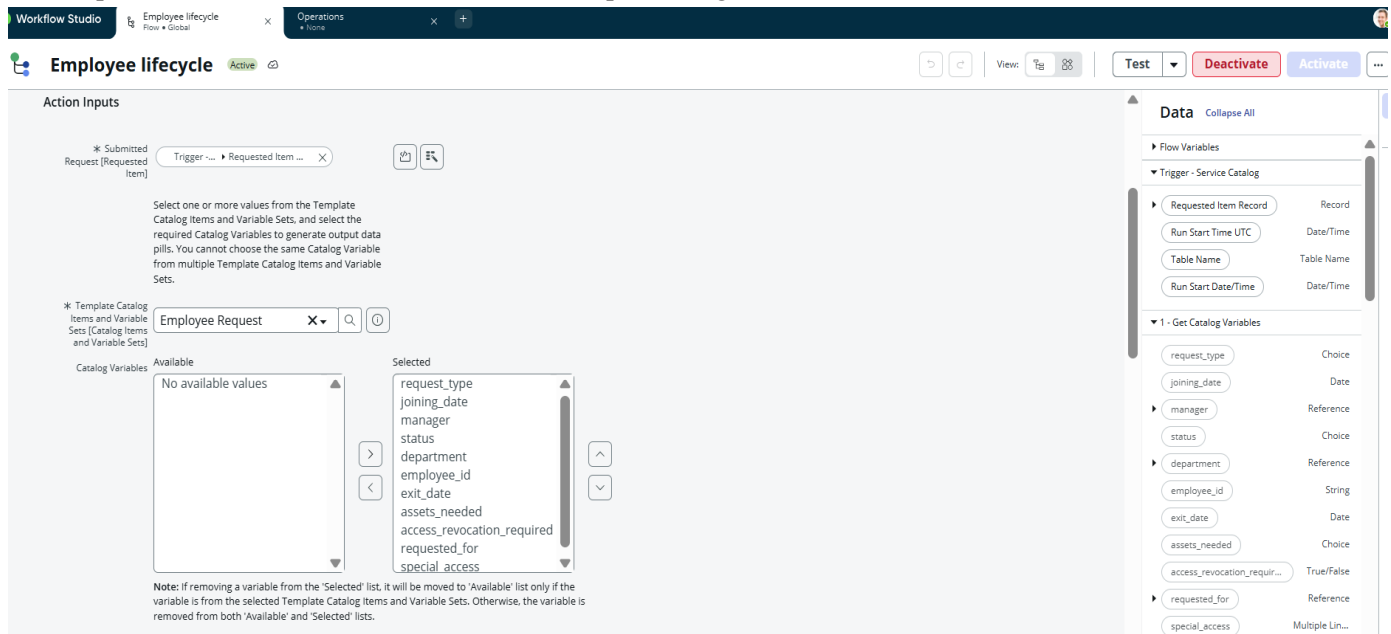
Step 3: Flow Designer – Initial Setup

A new flow was created in Flow Designer triggered when the catalog item is submitted.



Step 4: Get Catalog Variables Action

This step fetches the submitted variables for further processing.



Step 5: Create Lifecycle Record

A record is created in custom table:

Employee Lifecycle

(u_employee_lifecycle) This helps track employee request status.

Workflow Studio Employee lifecycle Flow Global Operations None

Service Catalog

ACTIONS Select multiple

- Get Catalog Variables from Employee Request from Employee Request
- Create Employee Lifecycle Record

Action Properties

Action Create Record

Action Inputs

* Table Employee Lifecycle [u_employee...] Data Pill Picker for Table employee...

* Fields

Employee ID	1 - Get Catalog V... ▶ departm...
Department	1 - Get Catalog Variab... ▶ stat...
Status	1 - Get Catalog Vari... ▶ mana...
Manager	1 - Get Catalog... ▶ requested...
Requested for	1 - Get Catalog V... ▶ joining_d...
Joining date	

+ Add field value

Data Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record

Run Start Time UTC	Date/Time
Table Name	Table Name
Run Start Date/Time	Date/Time

1 - Get Catalog Variables

request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

Step 6: Ask for Approval

Approval is sent to:

Manager (dynamic reference from form)

The request proceeds only after approval.

Workflow Studio Employee lifecycle Flow Global Operations None

Employee lifecycle Active

2 Create Employee Lifecycle Record

3 Ask For Approval on Requested Item

Action Properties

Action Ask For Approval

Action Inputs

* Record Trigger ... ▶ Requested Item ...

Table Requested Item [sc_req_item]

Approval Reason

Approval Field Approval

Journal Field Approval history

* Rules

Approve When: Anyone approves 1 - Get Catalog Vari... ▶ mana...

Due Date None

Add another OR rule set

OR AND

Delete Cancel Done

Data Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record

Run Start Time UTC	Date/Time
Table Name	Table Name
Run Start Date/Time	Date/Time

1 - Get Catalog Variables

request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

2 - Create Record

Step 7: Conditional Flow Logic

If Approved:

1. Create Catalog Tasks:
 - IT Task → Account setup & hardware
 - Facilities Task → Workspace preparation
 - Security Task → ID & access
2. Wait until:
All tasks are Closed Complete
3. Update records:
 - Lifecycle → Completed
 - RITM → Closed Complete
4. Send completion email.

The screenshot shows the 'Employee lifecycle' workflow in Workflow Studio. The 'If' condition is configured with the following settings:

- Condition Label: (empty)
- Condition 1: 3 - Ask For Ap... → Approval St... is Approved
- Buttons: Delete, Cancel, Done

The right sidebar shows the 'Data' panel with the following variables:

Variable	Type
request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

If Rejected:

- Update Lifecycle → Rejected
- Send rejection email to requester

The screenshot shows the 'Employee lifecycle' workflow in Workflow Studio. The 'Else' branch is configured with the following steps:

13. Else
14. Update Requested Item Record
15. Update Requested Item Record
16. Send Email

The right sidebar shows the 'Data' panel with the following variables:

Variable	Type
request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

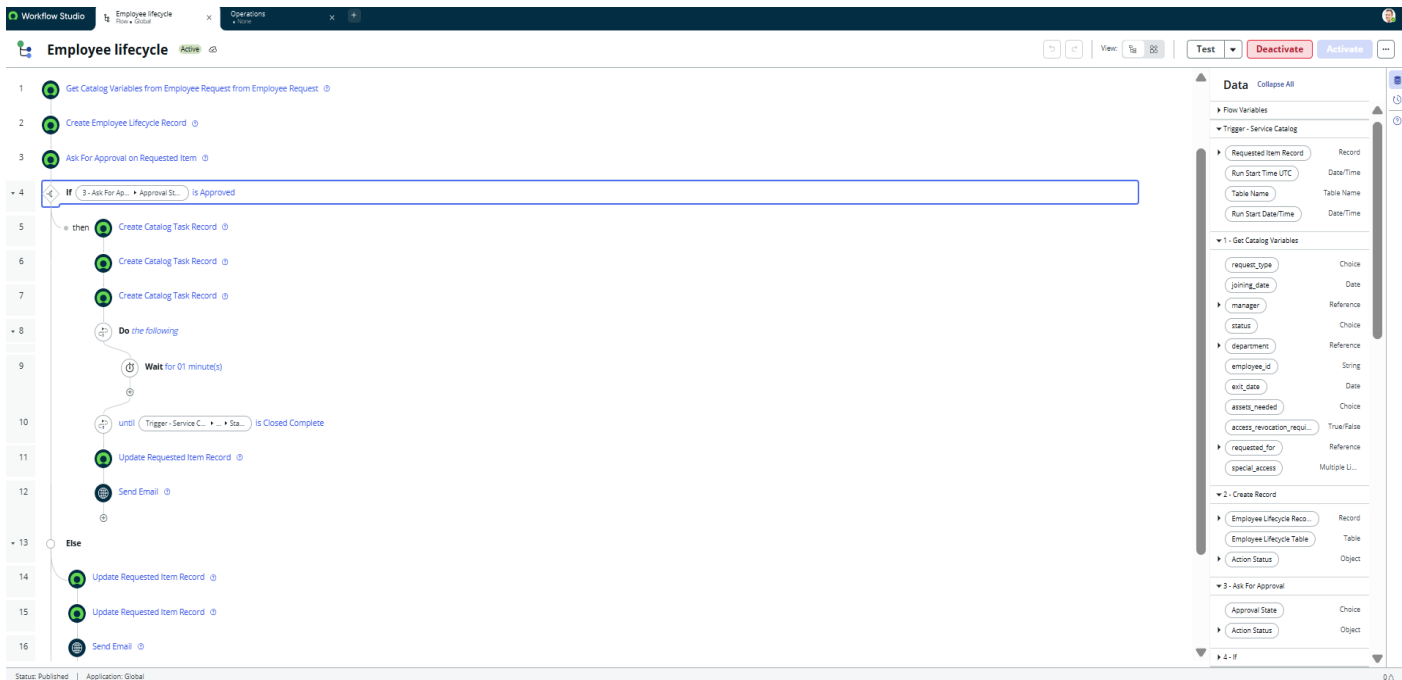
Step 8: Wait Condition Logic

The flow uses a loop mechanism to wait until:

All sc_task records reach Closed Complete

This ensures proper coordination between departments.

Step 9: Final Flow Design



The complete workflow is now automated end-to-end.

Results

1. Users can submit onboarding/offboarding requests easily
2. Requests are automatically processed and tracked
3. Tasks are assigned to relevant departments
4. Approvals and notifications are automated
5. Manual effort is significantly reduce

Conclusion

This project successfully demonstrates the automation of employee onboarding and offboarding using

ServiceNow. By integrating Service Catalog, Flow Designer, Approvals, and Notifications, the system ensures:

- Faster processing
- Improved coordination
- Better visibility
- Reduced manual errors

This solution can be extended to other HR and IT service processes.

Future Enhancements

1. Integration with Active Directory for account creation
2. SLA tracking for task completion
3. Virtual Agent (chatbot) integration
4. Dashboard for HR analytics and reporting

References

1. ServiceNow Documentation – <https://docs.servicenow.com>
2. GitHub Project Repository – <https://github.com/Priyahoskote/ServiceNow>